3D Model of Teapot Using Photogrammetry: Project Report

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1 Introduction

This project combines photogrammetry, computer graphics principles, and 3D rendering applications such as Blender to produce a 3D model and animation of a porcelain teapot. Photogrammetry, the process of converting physical objects into a 2D or 3D model, is a popular field within computer graphics and is widely used in many real-world tasks. This project used Blender and photogrammetry to create a 3D digital model of a teapot, demonstrating how photogrammetric technology can represent physical objects. The teapot's body, which is shaped like an octagon, is white with blue polka-dotted hearts. The handle and rim of the pot feature a gold trim, while the lid is blue and gold. In the scene, the teapot sits on a wood surface, which mimics a wooden table or desk. The animation displays the object from all angles, rotating both the teapot and the camera, lifting it, and tilting it to demonstrate a pouring motion.

The scope of this project is to show how the principles of photogrammetry and animation can merge to digitally represent a real-world object. Photogammetric technology has significantly impacted computer graphics and computer science as it has allowed experts in all fields to visualize physical scenes or objects virtually and modify their properties without affecting the real object. This is particularly useful in areas such as historical preservation, architecture, virtual reality and video games, the medical industry, and essentially anything where a virtual model might be useful. While simple, this project demonstrates some of the many possibilities of of photogrammetry and animation.

2. Literature Review

There is substantial research on the topic of photogrammetry. Often, experts combine photogrammetry and computer vision to better model and optimize real-word scenes and objects. For instance, this technology can be useful in cultural preservation, as discussed by Italian researchers Alessio Calantropio and Filiberto Chiabrando. In their article entitled "Underwater Cultural Heritage Documentation Using Photogrammetry" (2024), they explore applications and limitations of photogrammetric technology in underwater cultural heritage preservation. They claim it is a crucial resource for conservationists and historians as it "can measure distances, areas, and volumes of objects and features in a marine environment. These data can be used to assess the conditions of marine structures, track changes over time, and develop conservation plans" (Calantropio and Chiabrando 16). With this data, experts can preserve information about historical underwater sites, develop restoration strategies (potentially using computer vision), provide educational opportunities, and analyze human impact. However, accurately modelling underwater structures is often challenging due to "the different optical properties of water, light penetration, visibility and suspension, radiometric issues, and environmental drawbacks" in underwater modelling (Calantropio and Chiabrando 1). Their research discusses enhancements of current technology to overcome these difficulties. For example, due to the low-visibility of many underwater sites, the use of "high resolution cameras and strobe lighting, wide-field-of-view cameras, image stabilization and motion compensation, and image preprocessing techniques, such as noise reduction and denoising. . . can further improve image quality and reduce the impact of low-visibility conditions" (Calantropio and Chiabrando 3). Additionally, accurate calibration, dewarping algorithms, stereo reconstruction, and noise

reduction techniques can help to overcome optical distortions and noise that would obstruct a clear image (Calantropio and Chiabrando 11). Drones, which can capture aerial images of structures in shallow water, and remotely operated vehicles (ROVs), which "can operate in hazardous or inaccessible environments," also can provide higher resolution photos for a photogrammetric process more efficiently and safely (Calantropio and Chiabrando 14). Thus, while photogrammetry has significantly impacted historical preservation work, scientists continue to develop ways to enhance the technology to produce clearer and more accurate models.

Virtual reality (VR) and video games also significantly rely on photogrammetry. Exploring the integration of VR and photogrammetry, a group of researchers from Poland and the UK developed a system called PhotoTwinVR, which provides "an immersive gesture-controlled system for inspection, and dimension measurements of 3-D photogrammetric models of real-life structures in VR" (Tadeja et al. 143). Essentially, this system allows users to interact with 3D photogrammetric models of structures or objects without being physically near the real-world object. This is particularly useful for engineering surveys of structures such as pipe racks as it eliminates potential safety hazards, pedestrian closures, and the need for extra surveying materials. Using Function Analysis System Technique (FAST), the researchers created the system to accomplish four main tasks. First, "[t]he system should allow the user to visualize and interact with any given model representing a 3-D object" (Tadeja et al. 144). Second, the user can manipulate the model by rotation, displacement, and resizing. Third, the system should allow for visual model inspection. Finally, the system can take and store measurements of the 3D objects. To interact with the 3D models, users simply provide gestural input, as the system contains a hand-tracking component. To optimize user experience, "[t]he position of the hands

and their gestures are always signaled to the user using virtual hand representations" (Tadeja et al. 145). Users may also use a virtual toolbox to mark and manipulate the objects simply by moving their hands. After developing the system, the researchers conducted a study to determine its applicability. In the study, engineers were asked to "measure—using the interaction methods presented—the height of the supporting structure, and the distance between neighboring structures, as well as the width of one of them. Please take all other measurements that you consider relevant when surveying the asset" (Tadeja et al. 149). After completing the tasks, the participants were largely satisfied with the system despite some lack of detail and confusion caused by the "snapping grid" tool (Tadeja et al. 150). Overall, despite these minor flaws, PhotoTwinVR demonstrates how photogrammetry and VR may interact and assist engineers and architects in various structural assessments.

Photogrammetric technology may also aid in the digital reconstruction of urban scenes. Modelling large-scale scenes has many uses, "such as urban planning, navigation, virtual reality, and tourism" (Han et al. 7458). Thus, the optimization of this technology would significantly impact the usability and accuracy of these applications. However, 3D modelling of such large-scale scenes can be challenging due to their large size and complexity. Thus, recognizing these difficulties, researchers developed an "efficient multistep modeling framework for large-scale urban scenes from aerial images" (Han et al. 7458). Taking a 3D photogrammetric model and aerial images as input, the framework progresses through three major stages to create an urban scene reconstruction. First, using deep learning geometry data, the algorithm would segment the scene into various sections: "ground, building, [and] vegetation" (Han et al. 7461). Then, for each pixel in the images, it determines "the possibility of a pixel. . . belonging to one of the three categories. . . and the label corresponding to the maximum possibility is assigned to the

pixel category" (Han et al. 7461). Then, the program uses ray-tracing to project each type of pixel onto the appropriate area of a 3D scene. Second, the framework would examine each identified building and generate an orthographic image for it. Then, it detects the line segments of each building and divides "the ground plane into polygon cells" under each building (Han et al. 7460). Finally, using the polygons, the framework creates a simplified 3D model of each building. After testing it, the researchers found the framework produced "vectorized building models from photogrammetry meshes accurately and efficiently" (Han et al. 7469). This developed framework again shows the significance of photogrammetry in real-world applications, demonstrating its use in creating 3D models of urban scenes.

In conclusion, photogrammetry is a well-studied and continuously developing field, having many applications in areas such as historical preservation, virtual reality and structural evaluation, and urban scene reconstruction.

3. Methodology

3.1. Creating a Mesh

This project uses photogrammetry to digitally reconstruct a porcelain teapot. While I had first decided to model a teacup and saucer on a metal stand, I ran into issues when trying to create a mesh with the 3D reconstruction application Meshroom. After taking hundreds of photos from all angles of the teacup, Meshroom produced a discolored mesh filled with holes. Thus, I opted to try a different approach. Using the mobile app PolyCam, I scanned and took over 100 photos of the porcelain teapot. PolyCam created a much more accurate and cleaner mesh, which I imported into the 3D rendering application Blender.

3.2. Refining the Mesh

After importing the teapot mesh into Blender, I used the edit function to remove the table surface and any other unnecessary geometry. To refine the mesh, I used the decimate function to reduce its face count, which optimizes it and reduces computing power. I then used sculpting and painting tools to recolor, shape, and smooth the teapot, making it more realistic. Connecting an image texture node allowed me to paint over the current colors, refining the smaller details, such as the gold trim and blue hearts. Because PolyCam did not generate the base of the teapot, I used the fill function to fill in the gap and then used the paint function to make it more closely match the pot's base color. I also adjusted elements such as the metallic factor to make it appear more realistic.

3.3. Creating the Scene

Since the pot appeared dark from many perspectives, I added a few sun and area lights to the scene, ensuring its visibility from all angles. The pot sits on a surface plane, which I textured to appear as a wooden surface. I also have an image of the sky as the background to enhance its visual appeal.

3.4. Creating the Animation

To create the animation, I added keyframes where I wanted a specific motion. In the animation, the pot first rotates along the x-axis, displaying all its sides. Then, it lifts off the ground and tilts in a pouring motion. When it returns to the plane, the camera rotates 360 degrees around the object. The pot then lifts off the plane again and is translated along the sides of the

surface until it returns to its starting point. After rotating in the air along multiple axes, the pot returns to the origin, where it scales along all three axes before shrinking to its normal size.

4. Results

This section will provide images of the project's various steps and final results.

Figure 1: An image used in the initial PolyCam scan

Figure 2: An image used in the initial PolyCam scan

Figure 3: An image used in the initial PolyCam scan

Figure 4: Initial PolyCam mesh

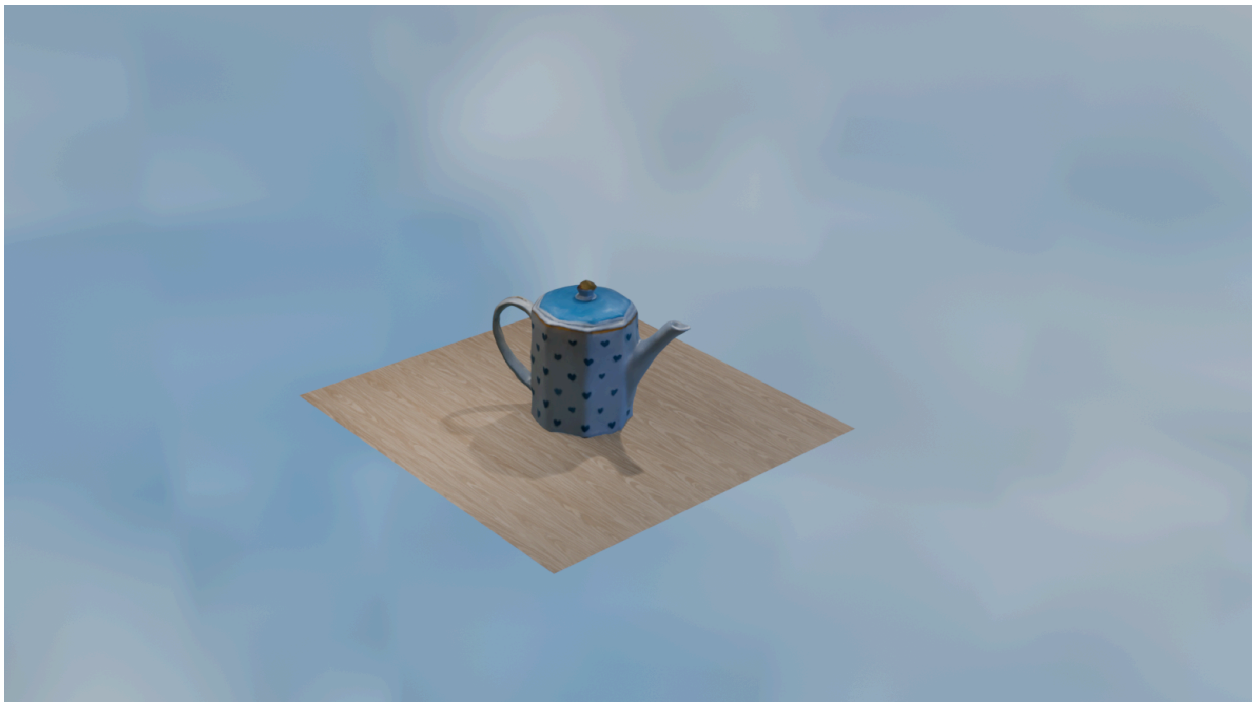

Figure 5: Final Blender render

The final Blender file and link to the animation, uploaded on YouTube, can be found in the GitHub repository: https://github.com/adavey5000/CSC322-Final-Project

5. Discussion

While simple, this project demonstrates an application of photogrammetry and animation to a real-world object. This and similar 3D models might be useful for someone analyzing or designing teapots and other dishware, as it could allow the viewer to visualize potential designs and coloring or experiment with various techniques. Additionally, models like this one may be useful on a website designed for the sale or design of china dishware. This project contributes to the advancement of computer graphics research by showing how one might use an available photogrammetry or animation application to construct a 3D model. While creating a 3D digital representation of an object may seem daunting, this project might encourage those learning the fundamentals of 3D modelling and animation to make use of the available software and resources. In this way, the project might increase the popularity of 3D modelling and inspire creativity and innovative thinking.

However, with its simplicity comes a few limitations. The teapot is not perfect, and some blemishes and imperfections remain. For instance, there may be small bumps or jagged edges in some areas. Additionally, coloring may appear somewhat uneven in the hearts, the trim, or the base. The model is a simple design and only features what Blender can provide, meaning that it may lack functionality or realism in some aspects.

Future work might include creating visual models for a website or animating a video for YouTube. Photogrammetric models often appear realistic, making them more visually appealing

and professional. Thus, adding such representations and animations may enhance a website's or video's charm and allure, attracting customers or viewers. Additionally, video games rely on animation techniques and representations of real-world objects. Due to the popularity of video games among both the youth and adults, creating animations and models for a game may serve as a fruitful foundation for future work in photogrammetry.

6. Conclusion

In conclusion, photogrammetry is a growing field of computer science and has many applications across various areas of expertise, including historical preservation, virtual reality, and urban scene reconstruction. Given the technology's increasing applicability, this project explores and demonstrates the integration and use of photogrammetry and 3D animation using both PolyCam and Blender. Using digital images of a teapot, PolyCam constructed a 3D mesh of the object. Once in Blender, sculpting, painting, textures, and other editing tools and features enhanced the teapot and gave it a realistic appearance. Blender's animation tools allowed for a smooth yet brief visualization of the teapot, demonstrating movements such as rotations, translations, and scaling. The project's simplicity and practicality allow it to demonstrate an example of the potential scope or foundation of 3D modelling and animation to those learning the principles of computer graphics and photogrammetry, encouraging ingenuity and imagination. Thus, despite its limitations, this project displays a simple yet accurate and comprehensive model of a teapot and serves as a foundation for further work in 3D modelling and animation.

Works Cited

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